

NEPHROLOGY TEACHING SERIES

Hypertension

Outpatient approach and resistant HTN

Home BP data outweighs office readings. Never escalate from one office measurement.

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MS3/MS4 rotation teaching

Why it matters

Common, silent, modifiable — and still the leading modifiable CV risk factor.

WHAT THE CLINIC VISIT COVERS

Confirm with home / ambulatory BP

Target: < 130/80; < 120 in CKD / high CV risk

Lifestyle, then staged 3-drug ladder

Adherence + side effects first

Secondary causes in resistant HTN

CV risk: lipid, DM, smoking

End-organ: UACR, ECG (LVH), fundus

TEACHING PEARL

*Office BP is a hypothesis, not a diagnosis.
Check adherence, cuff size, home log before
adding drugs. Most 'resistant' HTN is
pseudo-resistant.*

Diagnosis and targets

Measurement technique drives both.

Category	SBP / DBP (office)	Home average	Action
Normal	< 120 / < 80	< 120 / < 80	Lifestyle; recheck annually
Elevated	120–129 / < 80	120–129 / < 80	Lifestyle; recheck 3–6 mo
Stage 1	130–139 / 80–89	≥ 130 / ≥ 80	Lifestyle ± drug if CV risk ≥ 10%
Stage 2	≥ 140 / ≥ 90	≥ 135 / ≥ 85	Drug + lifestyle; 2-drug combo if ≥ 160/100
Hypertensive crisis	≥ 180 / ≥ 120	—	Urgent evaluation — end-organ damage? IV if yes

Targets by population: General < 130/80 (ACC/AHA) • CKD SBP < 120 (KDIGO 2021, automated office blood pressure (AOBP)-measured) • Elderly consider tolerability • Orthostasis → aim at standing SBP too.

Measurement — the most common error

Technique changes SBP by 10–20 mmHg. Do it right before escalating.

OFFICE BP

5 minutes quiet seating; feet flat on floor; no caffeine / nicotine / exercise 30 min prior

Correct cuff size: bladder width 40%, length 80% of arm circumference

Back supported, arm at heart level, uncovered

No talking during measurement

Average 2 readings at least 1 min apart

Automated office BP (AOBP) = unattended, automated; most aligned with SPRINT

HOME / AMBULATORY BP

Gold standard for diagnosis & titration

7 days of morning + evening readings, duplicates, discard day 1

Validated upper-arm cuff (not wrist)

Home BP averages are 5 mmHg LOWER than office — use HBP thresholds

White-coat HTN: elevated office, normal home → monitor only

Masked HTN: normal office, elevated home → treat

The drug ladder

Three classes, titrated to target. Add spironolactone if resistant.

Step	Agent	Preferred when	Avoid when
1	ACEi or ARB (not both)	CKD (UACR > 30), DM, HFrEF, post-MI	Pregnancy; bilateral renal artery stenosis (RAS); hyperK
2	Thiazide (chlorthalidone 12.5–25 mg preferred)	Older patient, Black patients, low-renin HTN	eGFR < 30 (use loop instead), gout
3	DHP calcium channel blocker (CCB) (amlodipine 5–10 mg)	Elderly isolated systolic HTN, Black patients	HFrEF (non-DHP), bradyarrhythmia (non-DHP)
4	Spironolactone 25–50 mg (or finerenone in DKD)	Resistant HTN on max 3-drug combo (PATHWAY-2)	K > 5.0, severe CKD; watch gynecomastia
+	Beta-blocker, clonidine, hydralazine	Add after 4th drug if still not controlled	Assess for secondary HTN at this point

Secondary hypertension — who to work up

Not a shotgun panel. Pick the test based on the clue.

Clue	Suspect	Test
Hypokalemia (spontaneous or diuretic-exaggerated)	Primary aldosteronism	Aldosterone/renin ratio (morning, off mineralocorticoid receptor antagonist (MRA) 4 wk)
HTN in young adult or resistant, with renal asymmetry	Fibromuscular dysplasia / atherosclerotic RAS	Duplex renal US → CTA/MRA
Flash pulmonary edema, acute Cr bump on ACEi	Bilateral RAS	Renal duplex / MRA
Paroxysmal HTN + headache, sweating, palpitations	Pheochromocytoma	Plasma metanephrines (free, fractionated)
Moon facies, central obesity, striae, DM, hypoK	Cushing syndrome	24-h urinary free cortisol; low-dose dexamethasone suppression test (LDDST); 11 pm salivary
Snoring, daytime sleepiness, BMI > 30	Obstructive sleep apnea	Home sleep study → PSG
HTN in pregnancy	Preeclampsia, chronic HTN	Urgent OB; ACEi/ARB contraindicated
Young HTN + weak femoral pulses / rib notching	Coarctation	CXR, echo, CTA/MRA

Resistant hypertension workup

True resistant HTN / pseudo-resistant. Rule out the latter first.

PSEUDO-RESISTANT (check first)

- Non-adherence (the #1 cause)
- White-coat effect — confirm with home/ambulatory BP
- Sub-optimal drug dosing (not titrated to max)
- Wrong cuff size; improper technique
- Volume overload (high Na diet, inadequate diuretic)
- Interfering substances: NSAIDs, OCPs, pseudoephedrine, alcohol, stimulants, herbal

If ruled out → true resistant; now pursue secondary workup

TRUE RESISTANT HTN — STEPS

Definition: BP > 130/80 on 3 maximized meds (incl. diuretic); OR controlled on 4 meds

Step 1: add spironolactone 25 mg (PATHWAY-2: best in resistant HTN)

Check K⁺ at 2 weeks; titrate to 50 mg if tolerated

Step 2: work up secondary HTN comprehensively

Step 3: add β -blocker, central α agonist, hydralazine

Renal denervation (investigational, limited) — SPYRAL/RADIANCE trials

Consider direct-acting vasodilators, nitroprusside only in crisis

Landmark HTN trials

The RCTs that set modern targets and drug choices.

Trial	Year	Takeaway
ALLHAT	2002	Chlorthalidone as good as ACEi or CCB for HTN — cheaper first-line option.
HOPE	2000	Ramipril ↓ CV events in high-risk non-HTN patients — BP-independent benefit.
ACCORD-BP	2010	Intensive BP (< 120) in T2DM — no CV benefit, more AE. Called SPRINT into question.
SPRINT	2015	SBP < 120 (by AOBP) ↓ CV 25% and mortality 27% vs < 140 in non-DM high-risk.
PATHWAY-2	2015	Spironolactone > bisoprolol, doxazosin for resistant HTN. NNT = 4.
HOT	1998	DBP target: < 80 = < 85 = < 90 in most, but DM benefited from < 80.
SYST-EUR	1997	CCB reduced stroke in elderly isolated systolic HTN.
STEP	2021	Intensive (< 130) vs standard (< 150) in elderly Chinese — 26% ↓ CV events.

Common clinic mistakes

The errors that lead to under-control or over-treatment.

AVOID

- Escalating from one office reading
- Prescribing without home BP log
- Ignoring adherence
- ACEi + ARB (ONTARGET)
- Thiazide at eGFR < 30
- Blaming ACEi for 10% Cr rise
- Missing OSA, NSAIDs, decongestants

DO INSTEAD

- 7-day home log first
- Pill count + open adherence Qs
- Check cuff size + technique
- One RAAS blocker, max dose
- Loop if eGFR < 30
- Accept Cr ↑ up to 30% on ACEi
- Always ask about OSA + OTCs

Clinical case

Think it through — answer is on the next slide.

VIGNETTE

A 52-year-old woman with T2DM, BMI 33, snoring. On lisinopril 40, HCTZ 25, amlodipine 10, metoprolol 50 BID. Office BP today 152/92 (Korotkoff, repeat 150/90). Home BP log: 148/88 to 155/94 am, 140/85 pm. Adherent by pill-count. K⁺ 3.4, spontaneously.

QUESTION: Is this resistant HTN? What's the most likely secondary cause and next step?

Clinical case — answer

Key teaching points from the case.

ANSWER

Yes (uncontrolled on 3 appropriate agents incl. diuretic). Primary aldosteronism — spontaneous hypokalemia + resistant HTN. Check aldosterone/renin ratio (morning, off MRA 4 wk).

TEACHING

True resistant HTN requires 3 appropriately-dosed agents including a diuretic. Before adding a 4th drug, always work up secondary causes. Primary aldosteronism is the #1 secondary cause in resistant HTN, especially if spontaneous or diuretic-exaggerated hypokalemia. Testing: aldosterone/renin ratio AM, ideally off MRA × 4 weeks when safe. Other secondary causes: OSA (she snores!), RAS (younger women → FMD), pheo, Cushing, CKD, CoA. Next step after workup: add spironolactone 25 mg (PATHWAY-2 NNT 4 for resistant HTN). Also: weight loss, sleep study.

REFERENCES

Hypertension

Guidelines

ACC/AHA 2017 Hypertension Guideline (+ 2025 update)

KDIGO 2021 Blood Pressure in CKD

Endocrine Society 2016/2025 Primary Aldosteronism

AHA 2018 Resistant Hypertension Scientific Statement

AHA 2019 BP Measurement in Humans

Landmark Trials

ALLHAT — JAMA 2002

HOPE — NEJM 2000

ACCORD-BP — NEJM 2010

SPRINT — NEJM 2015

PATHWAY-2 — Lancet 2015

HOT — Lancet 1998

SYST-EUR — Lancet 1997

STEP — NEJM 2021

UpToDate (2026 access):

Overview of hypertension in adults • Choice of drug therapy in primary (essential) hypertension • Evaluation of resistant hypertension • Primary aldosteronism: screening and diagnosis • Ambulatory and home blood pressure monitoring